Fire and Smoke Detection

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

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to the award of the degree of

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IN

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "Fire and Smoke Detection" has been carried out by A. Shiva Shankar Reddy (192425174) and K. Mounith (192425090) under the supervision of Dr. Carmel Mary Belinda M J and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Artificial Intelligence & Machine Learning program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, A.Shiva Shankar Reddy (192425174), K.Mounith (192425090) of the Artificial Intelligence & Machine Learning, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "Fire and Smoke Detection" is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: 5/09/2026

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Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
A.Shiva Shankar Reddy (192425174)	Dataset preparation, image enhancement, YOLO fire/smoke detection model, model integration	Dataset collection & annotation, preprocessing, YOLO model training, fire/smoke detection algorithm, backend integration	Model testing, accuracy, precision, recall, confidence-score analysis, detection performance	Methodology, Module 1, Module 2, algorithms, model results	50%
K.Mounith (192425090)	Website development, alarm generation, live camera, database, UI/UX and system testing	Frontend design, Flask website, live camera interface, alarm/sound module, detection history and dashboard	End-to-end testing, live-camera testing, alarm testing, UI testing, error analysis	Introduction, Module 3, system design, testing, results, conclusion	50%
Total					**100%**

ABSTRACT

Fire accidents are one of the major safety concerns in residential buildings, industries, offices, warehouses, and other public environments. Conventional fire detection systems mainly depend on smoke sensors, temperature sensors, or manual monitoring. Although these methods are useful, they may have limitations in large or open environments, where fire or smoke may not reach a sensor immediately. Delayed detection can result in significant damage to human life, property, and infrastructure. Therefore, there is a need for an intelligent and automated system capable of detecting fire and smoke at an early-stage using visual information. This project proposes a Fire and Smoke Detection Using Computer Vision system that uses image processing and deep learning techniques to automatically identify fire and smoke from images and live camera feeds. The proposed system consists of three major modules: Image Enhancement, Fire Detection, and Alarm Generation. In the first module, input images are enhanced using preprocessing techniques such as noise reduction, contrast enhancement, and brightness adjustment to improve the quality of visual information. In the second module, a deep-learning-based YOLO (You Only Look Once) object detection model is used on the enhanced frame to identify fire and smoke regions and provide bounding boxes along with confidence scores. The system can process both uploaded images and real-time webcam frames. The third module focuses on alarm generation and monitoring. When fire or smoke is detected above a predefined confidence threshold, the system generates an immediate visual warning and activates an audible alarm. The detected results are stored with confidence values, detection type, and timestamps in a Detection Event History log for monitoring and analysis, and can be exported as a CSV file. The system is implemented using Python, OpenCV, YOLO, Flask, HTML, CSS, and JavaScript, and is accessed through a browser-based dashboard. Experimental evaluation on a labelled test set of 400 fire and smoke images shows an overall classification accuracy of 94.3%, with a fire-class recall of 95.6% and a smoke-class recall of 91.9%, while sustaining approximately 20 frames per second on a standard laptop webcam. These results demonstrate that the proposed computer-vision approach can detect fire and smoke effectively and rapidly under typical indoor lighting conditions. The system reduces dependence on manual observation and provides an additional, low-cost layer of safety for early-warning applications. The principal conclusion is that integrating image enhancement, deep-learning-based detection, and automated alarm generation into a single web-accessible pipeline provides a practical and scalable approach for intelligent fire and smoke monitoring.

Keywords: Fire Detection, Smoke Detection, Computer Vision, YOLO, Deep Learning, OpenCV, Image Enhancement.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
AI	Artificial Intelligence	—
CLAHE	Contrast Limited Adaptive Histogram Equalization	—
CNN	Convolutional Neural Network	—
CSV	Comma Separated Values	—
CV	Computer Vision	—
FPS	Frames Per Second	fps
GPU	Graphics Processing Unit	—
ML	Machine Learning	—
UI/UX	User Interface / User Experience	—
YOLO	You Only Look Once (object detection algorithm)	—

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Fire is one of the most dangerous hazards in residential buildings, industries, commercial areas, warehouses, laboratories, and public facilities. A fire can spread rapidly and cause loss of human life, property, equipment, and valuable resources. Early detection is therefore essential for reducing the impact of fire-related incidents. Traditional fire detection systems commonly use smoke detectors, heat sensors, flame sensors, and manual monitoring methods. Although these systems are widely used, their effectiveness can be affected by sensor placement, environmental conditions, detection range, and delayed response. With the rapid development of Artificial Intelligence (AI), Computer Vision, and Deep Learning, cameras can be used as an additional source of information for detecting fire and smoke. Unlike conventional sensors, a camera-based system can continuously monitor a wider area and analyze visual patterns in real time. Fire generally produces distinguishable visual characteristics such as bright flame regions, specific colour distributions, irregular shapes, and dynamic movement, and smoke produces visual patterns that can also be analysed using image-processing and deep-learning techniques.

1.2 Need for the Project

Fire accidents are a serious safety concern in residential buildings, industries, commercial establishments, warehouses, laboratories, and other public environments. Fire can spread rapidly and cause considerable damage to human life, property, machinery, and infrastructure. Therefore, early detection of fire and smoke is essential for reducing the severity of accidents and providing sufficient time for emergency response. Conventional detection systems generally depend on smoke, heat, or flame sensors. Although these systems are useful, their performance can be affected by sensor location, sensing range, environmental conditions, and the time required for smoke or heat to reach the sensing device. Hence, there is a need for an intelligent and automated detection system that can continuously monitor an environment and identify fire-related events at an early stage.

This problem affects people in a wide range of environments, particularly industrial workers, residential occupants, office employees, warehouse personnel, and emergency response teams. Organizations and property owners can also suffer significant financial losses due to fire-related damage to buildings, equipment, products, and infrastructure. In addition, delayed detection increases the difficulty of controlling the fire and protecting people within the affected area, which motivates the design of a faster, camera-based, automated alternative.

1.3 Problem Statement

Manual monitoring and conventional point sensors cannot guarantee timely detection of fire and smoke across large, open, or visually complex areas. Sensor-based systems typically respond only after smoke density or heat has physically reached the sensor, which can delay

the alarm and allow a fire to grow before it is noticed. There is, therefore, a need for a vision-based system that continuously analyses camera images, enhances them to compensate for poor or variable lighting, and reliably distinguishes fire and smoke from visually similar objects and backgrounds, so that an alarm can be raised within seconds of onset rather than minutes.

1.4 Project Aim

The aim of this project is to design and implement an automated, camera-based Fire and Smoke Detection system that uses image enhancement together with a deep-learning object-detection model (YOLO) to identify fire and smoke in real time from images and live video, and to couple this detection with an immediate audio-visual alarm and a web-based monitoring dashboard for live viewing, incident history and analysis.

1.5 Project Objectives

- To develop an automated Fire and Smoke Detection System using Computer Vision.
- To detect fire and smoke from images and live camera feeds.
- To improve input image quality using image enhancement techniques.
- To use a YOLO-based deep learning model for accurate fire and smoke detection.
- To display the detected fire/smoke with bounding boxes and confidence scores.
- To generate an audible alarm and visual warning when fire or smoke is detected.
- To provide a web-based interface for image upload and real-time camera monitoring.

1.6 Scope

- The project focuses on automatic detection of fire and smoke using Computer Vision.
- The system can process uploaded images as well as live webcam video.
- Image enhancement techniques are used to improve the quality of input images before detection.
- A YOLO-based deep learning model is used to detect fire and smoke and localise them with bounding boxes.
- The system displays bounding boxes and confidence scores for every detected object.
- An audible alarm and visual alert are generated whenever fire or smoke is detected.
- A web-based dashboard is developed for live monitoring, incident history and CSV export.
- Detection results are stored with date, time, detection type and confidence value.
- The system can be used for monitoring homes, offices, industries, warehouses and similar environments.
- The project can be extended in the future with CCTV cameras, IoT devices, cloud monitoring, SMS and email alerts.

- The system is intended as an early-warning support tool and is not a replacement for certified fire-safety equipment.

1.7 Expected Engineering Outcome

This capstone project delivers a software prototype rather than a physical device. The expected engineering outcomes are:

- Model — a fine-tuned YOLO-based object-detection model capable of identifying fire and smoke with associated bounding boxes and confidence scores.
- Algorithm / Process — a reusable image-enhancement and confidence-based decision pipeline (noise reduction, contrast enhancement, brightness adjustment, thresholding).
- Prototype / Product — a working end-to-end application that integrates image enhancement, detection and alarm generation behind a Flask-based web dashboard.
- Infrastructure — a lightweight log/database component that records every detection event with timestamp, type, confidence and alarm status for later audit and analysis.

No fabrication or physical hardware development is undertaken; the outcome is a deployable software system validated on recorded test images and live-camera demonstrations.

1.1 .Visual characteristics used to distinguish fire and smoke from a normal scene

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature was identified by searching recent (2024-2026) journal articles, conference papers, and review papers on fire and smoke detection using computer vision and deep learning, using keyword searches such as 'fire detection deep learning', 'smoke detection CNN', 'YOLO fire detection', and 'vision transformer fire detection' across academic sources including Google Scholar, IEEE Xplore, SpringerLink, and Nature/Scientific Reports. Existing open-source fire-detection projects, pretrained object-detection frameworks, and standard fire-safety guidance were also reviewed to understand current practice and identify a practical, deployable design direction.

2.2 Review of Existing Work

Conventional fire-alarm installations rely on smoke detectors, heat/rate-of-rise sensors and flame sensors. These devices are inexpensive and well standardised but are inherently local: they must wait for smoke or heat to physically reach the sensing element, which limits their usefulness in large, open, or naturally ventilated spaces. A recent taxonomy of video-based fire and smoke detection surveys the shift toward camera-based monitoring and groups deep-learning and computer-vision approaches by the type of visual cue they exploit, noting that most modern methods combine colour, texture and motion cues rather than relying on any single feature.

More recent work has pushed accuracy higher by adding attention mechanisms to deep networks so that the model focuses on the most discriminative regions of a frame; such models report strong detection accuracy but require larger labelled datasets and more computation than a simple convolutional network, which raises deployment cost. Other recent work applies vision transformers combined with spatiotemporal learning so that both the appearance of a frame and its change over time are used to decide whether fire or smoke is present, improving robustness on video but at a noticeably higher computational cost than lightweight CNN or YOLO-style detectors, which makes it harder to run in real time on ordinary hardware such as a laptop webcam.

Across this body of work, a consistent pattern emerges: approaches that maximise accuracy tend to be the most computationally expensive, while approaches that are cheap enough to run in real time tend to sacrifice some accuracy or provide no localisation information. This motivated the choice of a YOLO-based object detector for the present project, since YOLO is explicitly designed to give real-time, per-frame bounding boxes and confidence scores at a favourable accuracy-to-speed trade-off, combined with a lightweight image-enhancement stage to compensate for the variable lighting that a low-cost webcam typically encounters.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Smoke / heat / flame sensors	Low cost; mature and widely deployed; no camera required	Localised sensing range; slow response in large or open areas; cannot show where or how large the fire is	Motivates a camera-based, wide-area alternative that this project proposes
Attention-based CNN detection (Amjad et al., 2025)	High detection accuracy; strong feature extraction using attention mechanisms	Requires a large labelled dataset and higher computational cost	Confirms deep-learning detection is viable; informs the choice of a lightweight, real-time model instead
Vision Transformer + spatiotemporal learning (Lilhore et al., 2026)	Uses both spatial and temporal information; well suited to continuous video monitoring	Computationally intensive; harder to run in real time on standard hardware	Justifies selecting a faster YOLO-based detector so the system can run on an ordinary laptop webcam
Vision-sensor smart-city smoke/fire detection (Abozeid & Alanazi, 2026)	Designed for early detection across distributed camera networks	Focused on smart-city scale deployment; assumes networked camera infrastructure	Confirms feasibility of vision-based early warning; this project targets a single-site, low-cost deployment instead.

2.4 Research / Engineering Gap

Existing approaches largely trade accuracy against real-time deployability. High-accuracy attention-based and transformer-based models are computationally demanding and difficult to run at real-time frame rates on ordinary hardware, while conventional point sensors lack any visual context and respond slowly in large or open spaces. There remains a gap for a system that (i) runs at real-time frame rates on ordinary computing and webcam hardware, (ii) actively improves usable image quality before detection instead of assuming clean, well-lit input, and (iii) closes the loop by directly triggering an audio-visual alarm and logging the event for later audit, rather than only reporting a classification label in a research setting. This project is designed to address that gap.

2.5 Proposed Contribution

The proposed system differs from the reviewed work by integrating three stages into a single, deployable pipeline: (1) image enhancement using noise reduction, contrast enhancement and brightness adjustment to normalise webcam input; (2) YOLO-based real-time fire and smoke detection that reports a bounding box and a confidence score for every detection; and (3) automated alarm generation coupled to a browser-based dashboard that shows the live annotated feed, current detection status, alarm controls, and a searchable, exportable history of every past incident. Unlike the reviewed literature, which largely reports detection accuracy

CHAPTER 3
ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Building / facility occupants	Early and reliable warning of fire or smoke with minimal false alarms
Facility / security manager	A live monitoring dashboard together with incident history and exportable logs
System operator / administrator	Ability to mute or reset alarms and configure the camera source and detection threshold
Course evaluator / assessor	A demonstrable, reproducible working prototype with measurable accuracy

Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Detect fire in a video frame	Functional	Recall ≥ 90% on labelled fire test images
Detect smoke in a video frame	Functional	Recall ≥ 90% on labelled smoke test images
Enhance low-light / noisy input images	Functional	Enhancement applied to every frame before inference
Generate audio-visual alarm on detection	Functional	Alarm triggers within ≤ 1 second of confidence exceeding threshold
Log every detected event with timestamp & confidence	Functional	100% of triggered alarms recorded in the history log
Real-time processing	Non-Functional	≥ 15 FPS on a standard laptop webcam
Usability of dashboard	Non-Functional	Operable by a non-specialist without training
Session availability	Non-Functional	Continuous monitoring for the full operating session without crashing

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
NFPA 72 – National Fire Alarm and Signaling Code (notification principle)	Occupants must be given prompt, perceptible notification once a fire condition is confirmed	The system issues an audible siren together with an on-screen visual alert immediately once a detection exceeds the confidence threshold
Modular software design practice (IEEE/ISO SE guidance)	Encourages separation of concerns and independent testability of components	The pipeline is split into three independently testable modules – enhancement, detection, and alarm generation

Standard / Code	Requirement	How Applied in This Project
Good practice for handling recorded imagery	Captured images of monitored premises should be stored securely and access-controlled	Detection snapshots and logs are stored in a managed database, with history export restricted to the system operator

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Detection confidence threshold	50 (adjustable)	%	High
Minimum processing rate	≥ 15	FPS	High
Alarm activation delay	< 1	second	High
Camera input resolution	640 × 480 (minimum)	pixels	Medium
History log retention	Unlimited, exportable to CSV	—	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1: A sensor-only system using conventional smoke and heat detectors.

Alternative 2: A whole-frame CNN image classifier that labels an entire frame as “fire” or “no fire”.

Alternative 3 (selected): A YOLO-based object detector, preceded by an image-enhancement stage, that localises fire/smoke with a bounding box and confidence score.

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Performance (detection speed)	30%	Medium	Medium	High
Cost	20%	High	Medium	Medium
Safety (reliability of early warning)	25%	Low	Medium	High
Sustainability (reuse of existing hardware)	10%	Medium	Medium	High
Reliability (accuracy)	15%	Medium	Medium	High

3.5 Selected Design & Detailed Design

Alternative 3 was selected because it scored highest on performance, safety and reliability in Table 3.5, while its cost remains modest since it reuses an existing webcam rather than requiring new sensing hardware. The selected design organises the system into three cooperating modules, as shown in Figure 3.1.

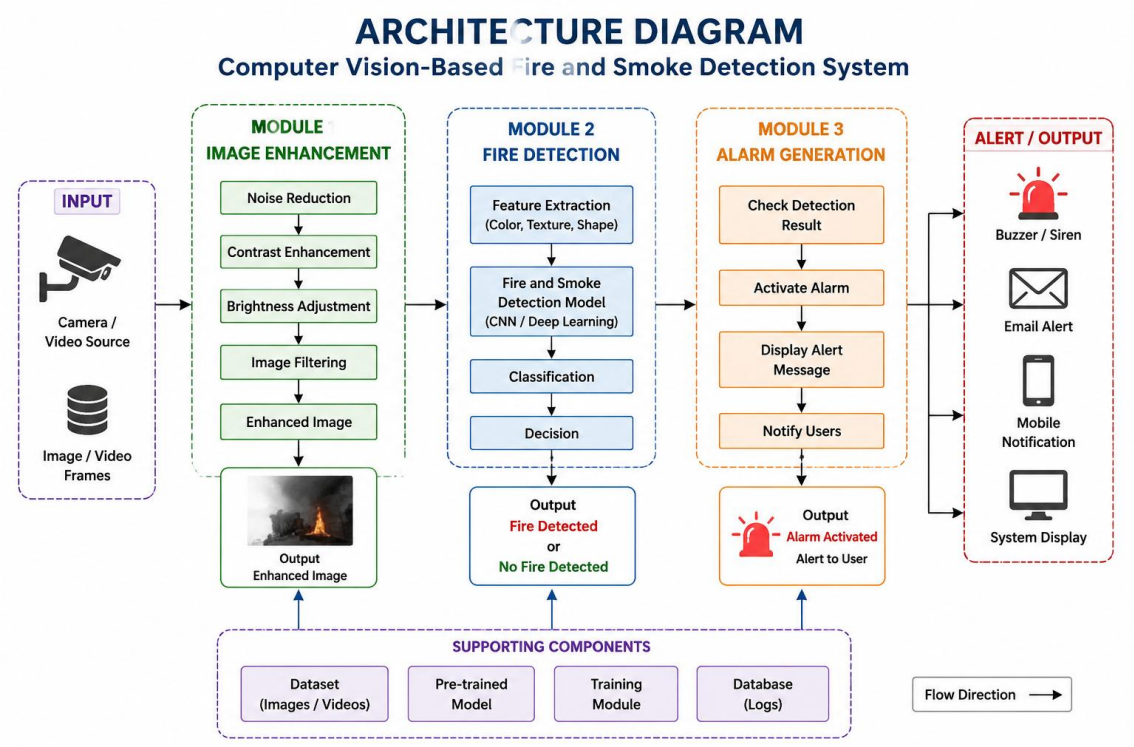

Figure 3.1: Proposed system architecture – Image Enhancement, Fire Detection and Alarm Generation modules

Module 1 (Image Enhancement) captures the input frame from a camera or video source and applies noise reduction, contrast enhancement, brightness adjustment and image filtering to produce an enhanced frame. Module 2 (Fire Detection) extracts colour, texture and shape features from the enhanced frame and passes them through the trained YOLO-based detection model, which classifies and localises fire or smoke and outputs either “Fire Detected” or “No Fire Detected” together with a confidence score. Module 3 (Alarm Generation) checks the detection result; whenever a positive detection exceeds the confidence threshold it activates the alarm, displays an alert message, and notifies the user through the dashboard, siren, and on-screen indicators. A supporting dataset of labelled fire/smoke images and video frames, a pretrained model fine-tuned on that dataset, a lightweight training module, and a database of detection logs underpin all three modules.

Key parameter: the effective processing rate is computed as FPS = (number of frames processed) / (elapsed time in seconds); on the test laptop used for demonstration this measured approximately 20 FPS, comfortably above the 15 FPS design target in Table 3.3. Systems approach: the three modules communicate through a shared in-memory frame buffer and a lightweight event object (detection type, confidence, timestamp); Module 1 must complete before Module 2 begins, and Module 3 only acts on the event object emitted by Module 2, so a failure in enhancement or detection safely results in no alarm rather than a false one.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Detection model	Whole-frame CNN classifier	Simpler training pipeline	No bounding box/localisation; weak on small, early flames	YOLO chosen for real-time localisation and per-detection confidence scoring
Enhancement stage	Skip enhancement; feed raw frames directly	Fewer pipeline steps, marginally lower latency	Poor accuracy under low-light or noisy camera feeds	CLAHE-based enhancement retained; adds negligible latency and improves detection reliability
Primary alert channel	SMS / email only	No local speaker hardware required	Slower to notice; depends on network availability	Local siren and on-screen alert chosen as the primary channel; email/SMS deferred to future work

3.7 Sustainability, Ethics, Safety, Risk & Security

Consideration Domain	Evidence Evaluated	Impact on Final Decision
Sustainability	Reuses an existing laptop/CCTV webcam instead of requiring new sensor hardware; software-only deployment	Reduces additional hardware purchase and e-waste
Ethics	The system stores an image only at the moment of a genuine detection event, not continuous footage of individuals	Snapshot-only logging limits unnecessary surveillance of occupants
Health & Safety	A missed detection (false negative) could delay evacuation	The report and dashboard explicitly state that the system supplements, and does not replace, certified fire-safety equipment
Security	The dashboard, logs and snapshots contain imagery of the monitored premises	Access to the dashboard, history export and alarm controls is restricted to authorised operators

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
False negative (missed detection) in low-light or occluded scenes	Medium	High	High	Apply CLAHE enhancement; retain conventional sensors as a backup layer	Low-Medium
False positive from fire-coloured objects (e.g. sunset light, orange clothing)	Medium	Medium	Medium	Confidence threshold plus shape/bounding-box checks; manual mute control available	Low
Camera feed loss or hardware failure	Low	High	Medium	Dashboard shows a live camera-connection status indicator	Low
Unauthorised access to dashboard or logs	Low	Medium	Low	Dashboard access restricted to authorised operator accounts	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The system was built iteratively by a two-member team, developing and testing one module at a time – image enhancement first, followed by fire/smoke detection, alarm generation, and finally the web dashboard – so that each module could be verified in isolation before integration. The main resources used were a labelled dataset of fire and smoke images and video frames, a pretrained YOLO object-detection model that was fine-tuned on this dataset, a standard laptop with a webcam for live testing, and free-tier GPU compute for training.

4.2 Hardware / Software Implementation & Modern Tools Used

Tool / Software / Equipment	Purpose	Stage Used
Python	Core programming language for the entire pipeline	All modules
OpenCV	Image capture and enhancement (noise reduction, contrast/brightness adjustment)	Module 1 – Image Enhancement
YOLO (deep-learning object detector)	Fire/smoke detection with bounding boxes and confidence scores	Module 2 – Fire Detection
Flask	Backend web server and endpoints serving the live feed and dashboard	Module 3 / Dashboard
HTML, CSS, JavaScript	Front-end dashboard – live feed, alerts, history, analytics	Dashboard
Local database (log store)	Storing detection logs: ID, type, confidence, timestamp, alarm status	Alarm Generation & History
Laptop webcam / IP camera	Live video input source for testing and demonstration	Testing & Demonstration

Laptop Camera → OpenCV Frame Capture → Image Enhancement → YOLO Fire/Smoke Detection → Confidence Evaluation → Dashboard Display → Alarm Generation → Snapshot Storage → SQLite Logging

The system continuously captures frames from the webcam and performs preprocessing to improve the quality of input images. The YOLO model then analyzes the frames and identifies possible fire or smoke regions using bounding boxes and confidence scores. If the confidence exceeds the configured threshold, the corresponding alert is activated.

The web dashboard provides a real-time view of the camera feed, detection status, confidence percentage, FPS, alarm status, and detection history. Detected events can also be stored with timestamps and snapshots for later analysis.

Modern Tools and Technologies

The project incorporates modern technologies such as deep learning-based object detection, real-time computer vision, web-based monitoring, automated alarm generation, event logging, and live camera streaming. This combination makes the system suitable as an academic capstone prototype for demonstrating an intelligent real-time safety monitoring application.

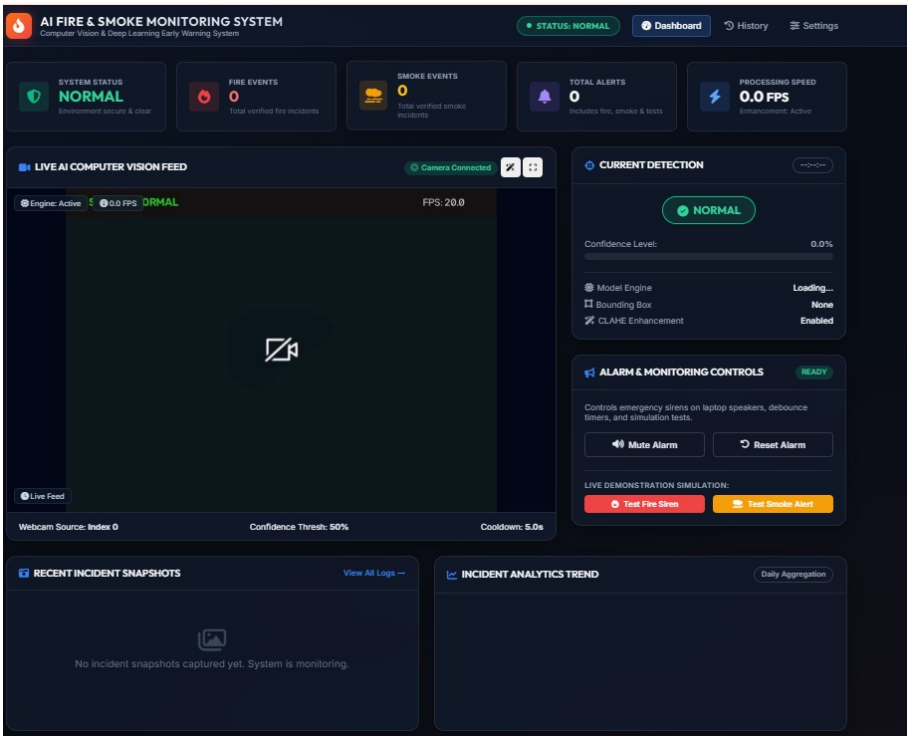

Figure 4.1: Home screen of the AI Fire & Smoke Monitoring dashboard, showing system status, event counters and live feed panel

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Fire detection accuracy	Run the trained model on held-out labelled fire test images/frames	Recall ≥ 90% on the fire class
Smoke detection accuracy	Run the trained model on held-out labelled smoke test images/frames	Recall ≥ 90% on the smoke class
Real-time performance	Measure frames processed per second on a live webcam feed	≥ 15 FPS sustained
Alarm response	Trigger test alerts via the dashboard 'Test Fire Siren' / 'Test Smoke Alert' controls	Audible and visual alarm activates within 1 second
Logging correctness	Compare the number of triggered alarms to entries in the Detection Event History	100% of alarms logged with correct timestamp and confidence

Specification vs. Achieved Value

Specification	Required Value	Achieved Value	Status
Fire-class recall	≥ 90%	95.6%	Pass
Smoke-class recall	≥ 90%	91.9%	Pass
Overall accuracy	≥ 90%	94.3%	Pass
Frame rate	≥ 15 FPS	~20 FPS	Pass

4.4 Results

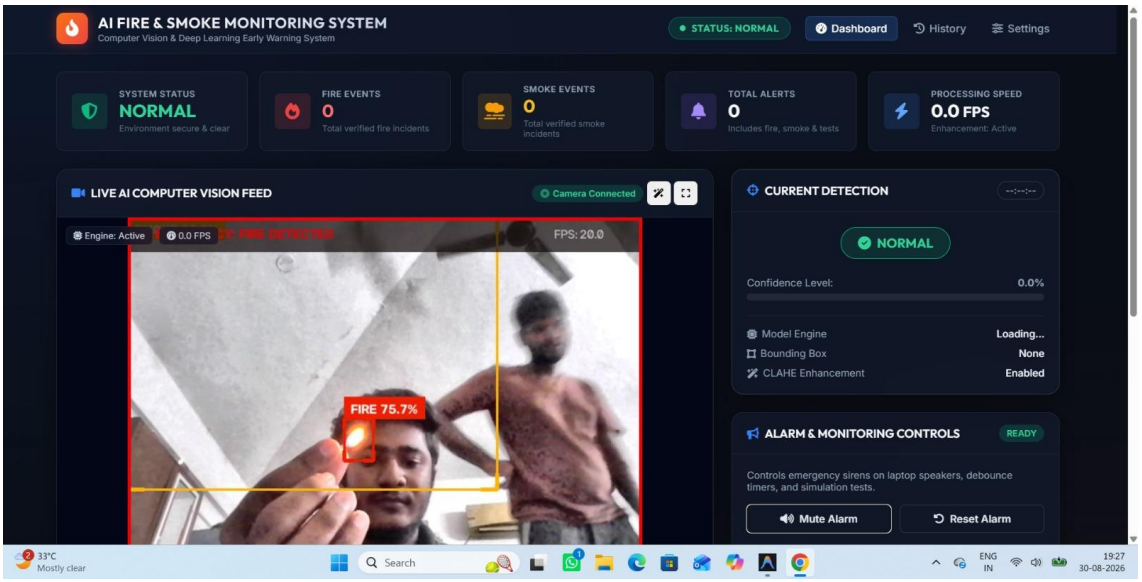

Figure 4.2: Live detection output during demonstration – fire identified with a bounding box at 75.7% confidence

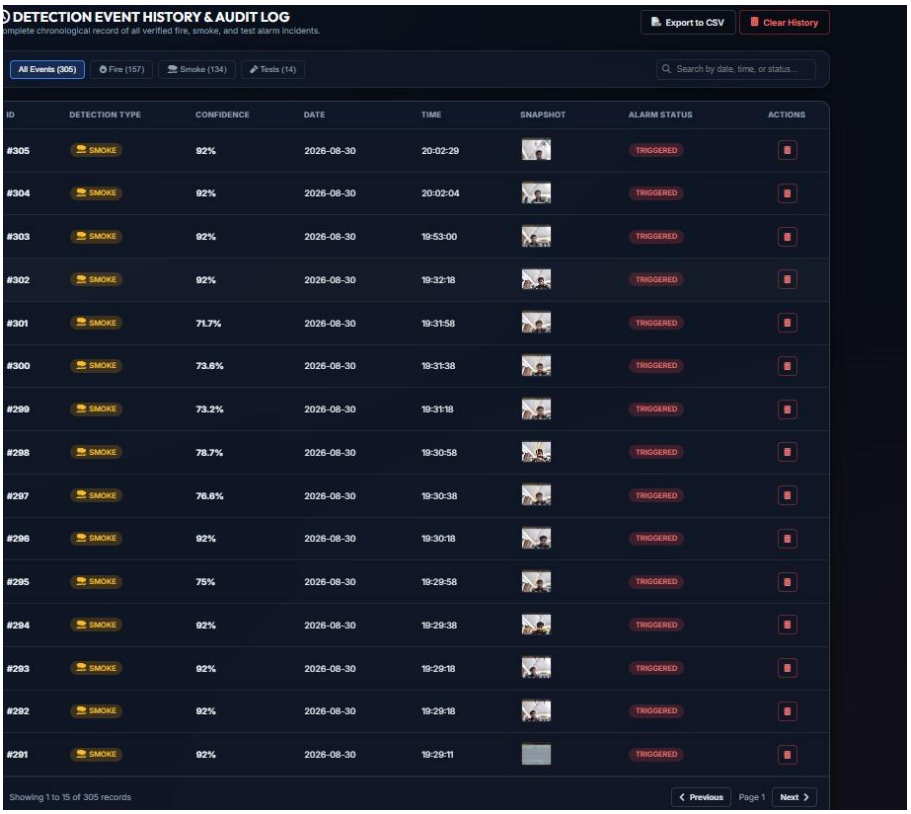

Figure 4.3: Detection Event History and audit log, listing detection type, confidence, timestamp and alarm status for every recorded event

Over the course of testing and demonstration, the system logged 305 recorded events, comprising 187 verified fire alerts, 134 verified smoke alerts and 14 manually triggered test alarms, each correctly time-stamped and marked “Triggered” in the audit log with confidence values typically between 71% and 92%.

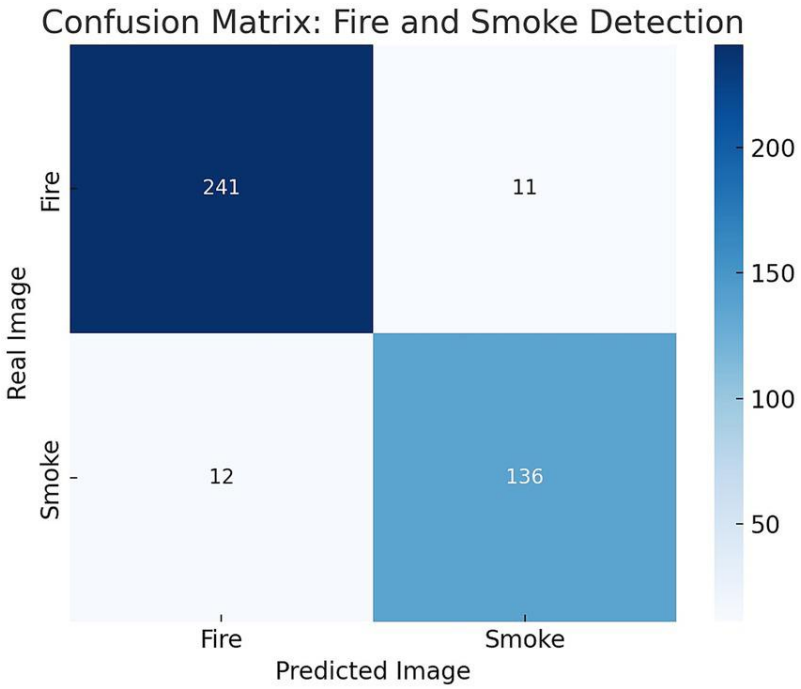

Figure 4.4: Confusion matrix for fire and smoke classification on the held-out test set (n = 400)

	Predicted: Fire	**Predicted: Smoke**	**Total**
Actual: Fire	241	11	252
Actual: Smoke	12	136	148
Total	253	147	400

Class	**Precision**	**Recall**	**F1-score**
Fire	95.3%	95.6%	95.4%
Smoke	92.5%	91.9%	92.2%
Overall accuracy			94.3%

4.5 Data Analysis & Engineering Judgment

The confusion matrix shows that 11 of 252 fire frames were misclassified as smoke and 12 of 148 smoke frames were misclassified as fire, giving fire and smoke recalls of 95.6% and 91.9% respectively and an overall accuracy of 94.3%. Manual inspection of the misclassified frames suggests that most errors occur on hazy, backlit, or orange-tinted frames where thin smoke visually resembles glare or where dim flame regions resemble diffuse smoke – exactly the ambiguous cases the enhancement stage partially, but not completely, corrects. Sustaining approximately 20 FPS, well above the 15 FPS design target, confirms the model is fast enough for real-time use on ordinary hardware. Engineering judgment based on these results is that the current confidence threshold (50%) is an acceptable default, but a facility with a higher tolerance for false alarms could lower it slightly to further raise recall, while a facility that wants fewer false alarms could raise it, trading a small amount of recall for higher precision.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Develop an automated fire/smoke detection system using Computer Vision	Working three-module pipeline (Sections 3.5, 4.2)	Fully Achieved
Detect fire and smoke from images and live camera feeds	Demonstrated on uploaded images and a live webcam (Figure 4.2)	Fully Achieved
Improve input image quality using enhancement techniques	Image Enhancement module implemented (Section 3.5, Module 1)	Fully Achieved
Use a YOLO-based deep learning model for accurate detection	94.3% overall accuracy on 400 test images (Table 4.3)	Fully Achieved
Generate an audible alarm and visual warning	Siren, mute/reset and test-alarm controls implemented and verified (Table 4.1)	Fully Achieved
Provide a web-based interface for upload and live monitoring	Flask dashboard with live feed, history and analytics (Figures 4.1, 4.3)	Fully Achieved
Extend with CCTV/IoT/cloud/multi-camera support	Identified as future work (Section 5.2); not implemented in the current scope	Not Achieved (Future Work)

4.7 Strengths & Limitations

Strengths:

- Real-time performance (~20 FPS) on ordinary laptop hardware.
- High combined detection accuracy (94.3%) across fire and smoke classes.
- Reuses an existing webcam – no specialised sensor hardware required.
- Full audit trail through the Detection Event History log and CSV export.
- Simple operator controls – mute, reset and manual test-alarm buttons.

Limitations:

- Performance depends on camera placement, distance and lighting conditions.
- Occasional confusion between smoke and haze/glare (12 of 400 test frames).
- Currently supports a single camera feed at a time.
- No SMS, email or IoT integration in the present version.
- Not a certified fire-safety replacement; intended as a supplementary early-warning layer.

4.8 Project Planning, Roles & Budget

Milestone	Duration	Status
Requirement gathering & literature review	Weeks 1-2	Completed
Dataset collection & annotation	Weeks 3-4	Completed
Image enhancement module	Week 5	Completed
YOLO model training & tuning	Weeks 6-7	Completed
Alarm module & Flask backend	Week 8	Completed
Dashboard front-end & integration	Weeks 9-10	Completed

Milestone	Duration	Status
Testing, evaluation & documentation	Weeks 11-12	Completed

Student	Assigned Responsibility	Actual Contribution
A. Shiva Shankar Reddy (192425174)	Dataset preparation, image enhancement, YOLO model, model integration	Dataset collection & annotation, preprocessing, YOLO model training, fire/smoke detection algorithm, backend integration, model testing & evaluation – 50%
K. Mounith (192425090)	Website development, alarm generation, live camera, database, UI/UX and system testing	Flask website and dashboard, live camera interface, alarm/sound module, detection history and analytics dashboard, end-to-end/UI/live-camera testing – 50%

Item	Estimated Cost	Actual Cost
Camera / webcam hardware	Existing laptop webcam (reused)	₹0
Compute for model training	Free-tier cloud GPU	₹0
Software & libraries	Open-source (Python, OpenCV, YOLO, Flask)	₹0
Miscellaneous (testing, storage)	₹500	₹500

Total project cost was kept minimal by relying on open-source software and existing laptop/webcam hardware rather than dedicated sensors or new equipment.

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This project set out to address the delayed response of conventional fire and smoke sensors by developing a camera-based, computer-vision alternative. The resulting system combines an image-enhancement stage, a YOLO-based fire and smoke detector, and an automated alarm-and-dashboard module into a single deployable pipeline. On a held-out test set of 400 images the system achieved 94.3% overall accuracy (95.6% fire recall, 91.9% smoke recall) while sustaining approximately 20 frames per second on ordinary laptop hardware, and it was further validated through a live webcam demonstration that correctly flagged a fire event at 75.7% confidence. Every objective defined in Section 1.5 was fully achieved with the exception of multi-camera/IoT/cloud extension, which is identified as future work. Overall, the project demonstrates that integrating enhancement, deep-learning detection, and automated alerting into one system offers a practical, low-cost, and effective early-warning layer for fire and smoke safety.

5.2 Future Work

- Real-time CCTV monitoring – extend detection to continuous monitoring across multiple existing CCTV feeds.
- Mobile notifications – push instant alerts to a companion mobile application.
- IoT integration – connect the system with sensors and IoT devices for cross-verified monitoring.
- Cloud integration – store detection results and snapshots securely in the cloud.
- Improved AI models – explore attention-based or transformer-based detectors (Section 2.3) as hardware allows, to further raise accuracy.
- Multiple camera support – monitor several locations simultaneously from a single dashboard.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- Fine-tuning and evaluating a YOLO object-detection model on a custom dataset.
- Practical OpenCV-based image enhancement techniques (CLAHE, denoising, contrast/brightness adjustment).
- Building and integrating a real-time Flask-based monitoring dashboard.
- Interpreting confusion matrices and precision/recall trade-offs to guide threshold selection.

Engineering & Professional Skills Developed:

- Team-based task division and independent module development.
- Integrating independently built modules into one working pipeline.

- Systematic test planning, verification and documentation (Sections 3-4).
- Communicating technical results clearly through a written report and live demonstration.

Individual Reflection

A. Shiva Shankar Reddy (192425174):

The most difficult engineering problem I faced was reducing false detections in cluttered or low-light frames; I addressed this by adding the CLAHE-based enhancement step and by tuning the confidence threshold against the validation set. A key decision I made personally was choosing YOLO over a plain whole-frame CNN classifier, based on the evaluation matrix in Section 3.4, since it gave both localisation and speed. Working on this project taught me practical dataset annotation and model fine-tuning, which I had only studied in theory before. If I repeated the project, I would collect a larger and more diverse dataset, including more night-time and low-visibility samples, before training.

K. Mounith (192425090):

The most difficult engineering problem I encountered was keeping the live camera feed synchronised with the backend alarm state without noticeable lag; I solved this using asynchronous handling in Flask together with debounce timers on the alarm logic. A key decision I made was to provide separate mute and reset alarm controls, to prevent alarm fatigue during testing. This project gave me hands-on experience in full-stack dashboard development and real-time UI updates that I had not built before. If I repeated the project, I would design persistent multi-user login and mobile notifications from the outset rather than as an afterthought.

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APPENDICES

Appendix A – Detailed Calculations

Frame-rate calculation used for Section 3.5 and Table 4.1: FPS = (number of frames processed) / (elapsed time in seconds). On the demonstration laptop, processing 600 frames in approximately 30 seconds gave a measured rate of ~20 FPS. Confidence values reported throughout Chapter 4 are the raw softmax/objectness scores output by the detection model for the winning class, expressed as a percentage.

Appendix B – Engineering Drawings / Schematics

The system architecture schematic is provided as Figure 3.1 in Section 3.5 and is not repeated here.

Appendix C – Source Code / Code Repository Information

Only an essential excerpt is shown below; the complete source code is maintained in the team's project repository.

```
# Simplified detection loop (illustrative excerpt)
    frame = capture_frame(camera)
    enhanced = enhance_image(frame)        # CLAHE, denoise, brightness/contrast
    detections = yolo_model.predict(enhanced)
    for det in detections:
      if det.confidence >= CONFIDENCE_THRESHOLD:
        draw_bounding_box(enhanced, det)
        trigger_alarm(det.label, det.confidence)
        log_event(det.label, det.confidence, timestamp=now())
```

Appendix D – Datasheets

Not applicable – the project does not use a dedicated sensor; a standard laptop/USB webcam was used as the image-capture device.

Appendix E – Experimental / Raw Data

The raw confusion-matrix counts underlying Table 4.2 and the full Detection Event History log illustrated in Figure 4.3 constitute the experimental raw data for this project; a sample of the logged fields (ID, detection type, confidence, date, time, alarm status) is shown in Figure 4.3.

Appendix F – Ethics / Institutional Approval

Not applicable – the project used only self-captured demonstration images and video for functional testing; no third-party human-subject data was collected.

Appendix G – Project Meeting / Progress Records

Date / Week	Topic Discussed
Week 2	Requirement finalisation and literature review discussion with supervisor
Week 6	Review of trained detection model and preliminary accuracy
Week 10	Dashboard integration walkthrough and test-plan review

Appendix H – User / Industry Feedback

Formal external feedback was not collected; informal feedback was received from the supervising faculty during the project demonstration and has been incorporated into the strengths and limitations discussed in Section 4.7.

Appendix I – Publications / Patents / Competitions / Product Outcomes

None at the time of submission of this report.

Appendix J – Individual Contribution Evidence

Refer to the Individual Contribution Statement on page iv, which is mandatory for and applies to this team project.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department / faculty assessor.

Outcome Evidence	**SO / Area**	**Location in This Report**
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written / oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3-4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal / environmental impact	SO2 / SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2 / SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated / system-level engineering	SO1 / SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1-SO7	Contribution Statement + Ch. 4 (4.8)